

The Editors
International Journal of Forecasting

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Dear Editors,

We are pleased to submit our manuscript, “**Online Anchor-Robust Forecasting under Economic Regime Shift,**” for consideration as a research article in the *International Journal of Forecasting*.

Economic and financial time series are punctuated by regime shifts — geopolitical shocks, monetary-policy turns, energy crises, regulatory interventions — and the dominant response in the online-learning and conformal-prediction literatures is *reactive*: detect that a change has occurred and then recalibrate. Such methods are always one step behind and pay a re-adaptation cost at every break. Our paper develops and evaluates a complementary, *proactive* strategy for the common case in which shifts travel through an observable channel of regime drivers, such as geopolitical-risk or policy-uncertainty indices, and pairs it with a practical answer to the objection such methods usually face: what happens when the channel is not handed to you correctly specified?

The paper makes four contributions. First, we bring *anchor regression* — until now a batch estimator — online, as OARF: a leak-free, first-order streaming learner that drives the component of its residual correlated with a driver channel toward zero using past-only exponential-moving-average moments, and that reduces exactly to online gradient descent when its robustness penalty is switched off. Second, because the channel is not always known in advance, we study an online channel-discovery add-on and show that its central failure mode — a minority of streams settling on a poorly identified channel and incurring a heavy-tailed worst-case cost — is *detectable without ground truth*: an eigengap diagnostic computed from the discovery layer’s own statistics tracks true recovery quality and, used as an automatic gate that falls back to the fixed-channel update, recovers roughly a quarter of the mean worst-case error the ungated learned channel gives up. Third, we propose *interventional regret* as an evaluation lens for online robustness and prove the fixed-channel learner is no-regret against a switching comparator under linear structural-causal-model assumptions, with two explicit rates for oracle and online moment estimation. Fourth, we provide a careful empirical study — a controlled SCM with a held-out intervention grid, plus sixteen years of public daily energy variance-proxy data — together with stress tests, assumption-violation scenarios and a pre-deployment checklist that map precisely *when* proactive anchor robustness helps and when it fails.

We have tried to be scrupulously honest about scope throughout. The method delivers large reductions in worst-case interventional error when the anchor is informative and shifts are anchor-mediated, but we show, and report plainly, that the advantage shrinks or reverses under noisy, misspecified or non-anchor-mediated shifts, and that the channel-discovery add-on remains outside our regret theorem even with the diagnostic gate. On the real energy data all methods lie within the Model Confidence Set, so OARF is competitive but not statistically superior, and we say so rather than over-reading the point estimates. We believe this combination — a principled streaming method, a validated safeguard against not knowing the channel, an interventional-robustness evaluation protocol, and a diagnostic account of the method’s limits — is a useful and publishable contribution at the interface of online learning, causal robustness and economic forecasting,

and is well suited to the *International Journal of Forecasting*'s focus on methodologically rigorous, practically evaluated forecasting research.

To support full reproducibility, all code, data-build scripts and the exact commands that regenerate every number, table and figure are publicly available at <https://github.com/vinhqdang/OARF-Online-Anchor-Robust-Forecasting-under-Economic-Regime-Shift>.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere; that all authors have read and approved the submission; and that we have no conflicts of interest to declare. The study uses only publicly available, non-personal data and required no ethics approval. Should the journal operate double-blind review, the repository above can be replaced with an anonymised mirror and the link withheld until acceptance.

Thank you for considering our work. We look forward to your response.

Sincerely,
The Authors